

Problem Approach

1.Objective:

The task is to utilize the YOLOv8 pose detection model to identify human poses, with a focus on extracting hand keypoints. The solution should produce both a visual representation (annotated image) and a JSON file containing the coordinates of the detected hands.

2.Step-by-Step Process:

Model Initialization:

- We start by loading the pre-trained YOLOv8 model for human pose detection using `YOLO('yolov8n-pose.pt')`. This model is trained to identify various body keypoints, including hands.

Image Input:

- The input image (which contains humans) is loaded using `cv2.imread()`. This will be the source for detecting keypoints.

Pose Detection with YOLO:

- The YOLO model processes the image and detects the pose keypoints of any humans in the image. Each human's body parts, including hands, are identified.

Hand Keypoint Identification:

- After detecting the pose, we focus on the hands. In the YOLO model's keypoints output, the left hand is typically found at index 9 and the right hand at index 10. We extract their `(x, y)` coordinates for further use.

Visual Annotation:

- To highlight the detected hand keypoints, circles are drawn on the image at the extracted coordinates. Green circles represent the left hand, and red

circles represent the right hand.

Saving Outputs:

- The modified image is saved with a new name, showing the detected keypoints.
- Additionally, the extracted hand keypoints are saved in a JSON format, structured like:

```
{
  "hands": [
    {
      "imagename": "image.jpg",
      "hand1": [x1, y1],
      "hand2": [x2, y2]
    }
  ]
}
```

3.Handling Edge Cases:

- If no hands are detected, the process ensures no erroneous data is included in the output.
- The solution also caters to images with multiple people, extracting hand keypoints for each detected individual.

4.Output:

Annotated Image: The image with visible markings (green and red) showing the left and right hand keypoints.

JSON File: A structured JSON file containing the coordinates of both hands for each detected person in the image.

5.Libraries Used:

- **Ultralytics YOLO:** Provides the ability to detect and extract pose keypoints, including hands.
- **OpenCV:** Handles reading, modifying, and saving images with annotations.
- **JSON:** Ensures the output is saved in a readable, structured format for easy use.

Summary:

This approach combines pose detection, hand keypoint extraction, and image annotation into a simple workflow. By leveraging YOLOv8 for pose detection and OpenCV for image manipulation, the solution provides both visual and data-driven results. The JSON output ensures that the coordinates are saved in an easily accessible format, making the solution robust and adaptable to different image scenarios.